CHEM 13 | ORGANIC CHEMISTRY LECTURE



"ONE MOLECULE, FOUR REPRESENTATIONS"

From Molecular Formula to Bond-Line Structure

Prepared by:
Kathly M. Alcantara
BSChE-2A





CARD 01

Propane

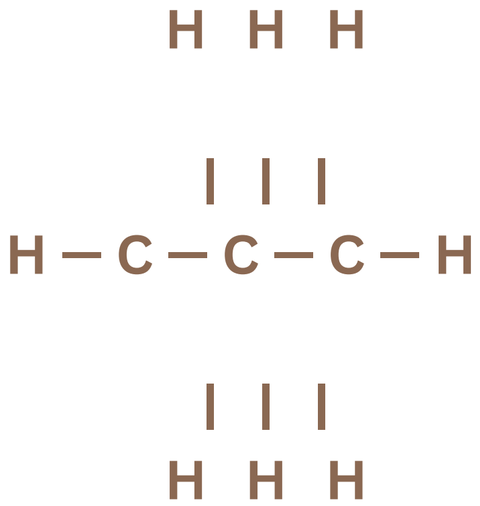
1.Molecular Formula

C_3H_8

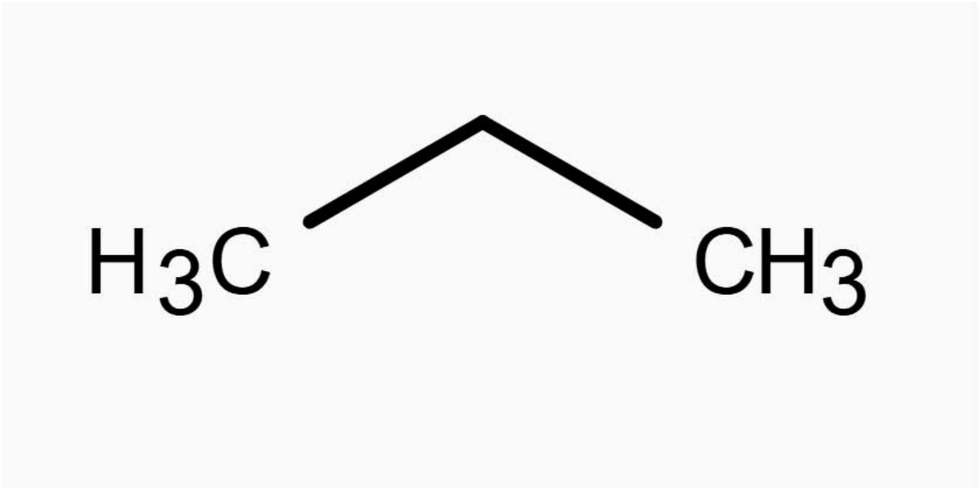
2.Condensed Structural Formula

$CH_3CH_2CH_3$

3.Lewis Structure



4.Bond-Line Formula



CARD 02

Cyclohexane

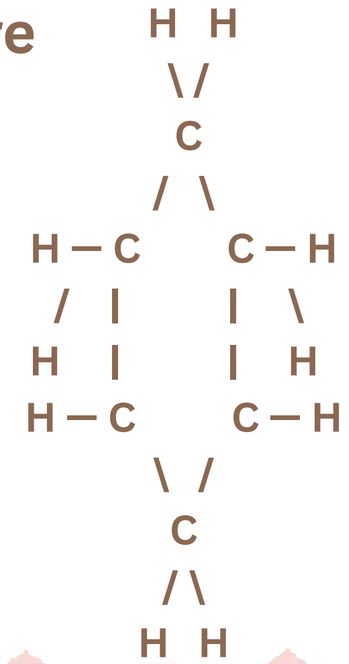
1.Molecular Formula

C_6H_{12}

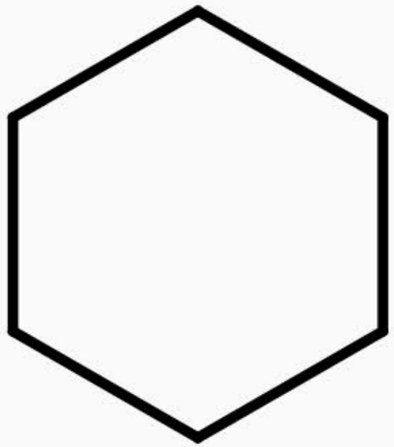
2.Condensed Structural Formula

C_6H_{12} (cyclic: cyclo-$(CH_2)_6$)

3.Lewis Structure



4.Bond-Line Formula



CARD 05

Ethanol

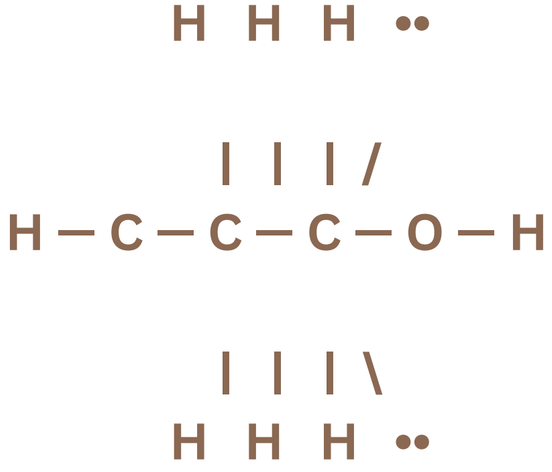
1.Molecular Formula

C_2H_6O

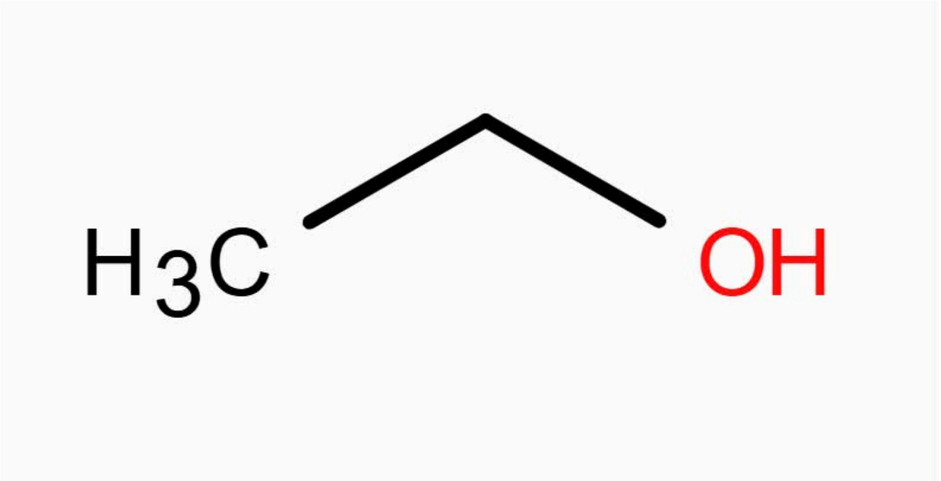
2.Condensed Structural Formula

CH_3CH_2OH

3.Lewis Structure



4.Bond-Line Formula



CARD 03

Acetic Acid

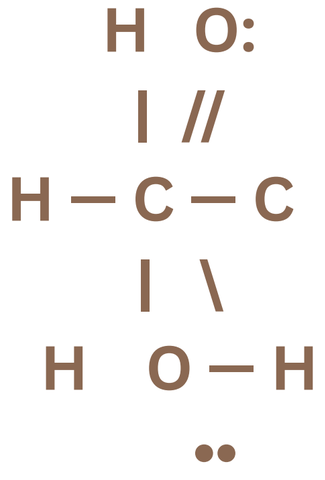
1.Molecular Formula

$C_2H_4O_2$

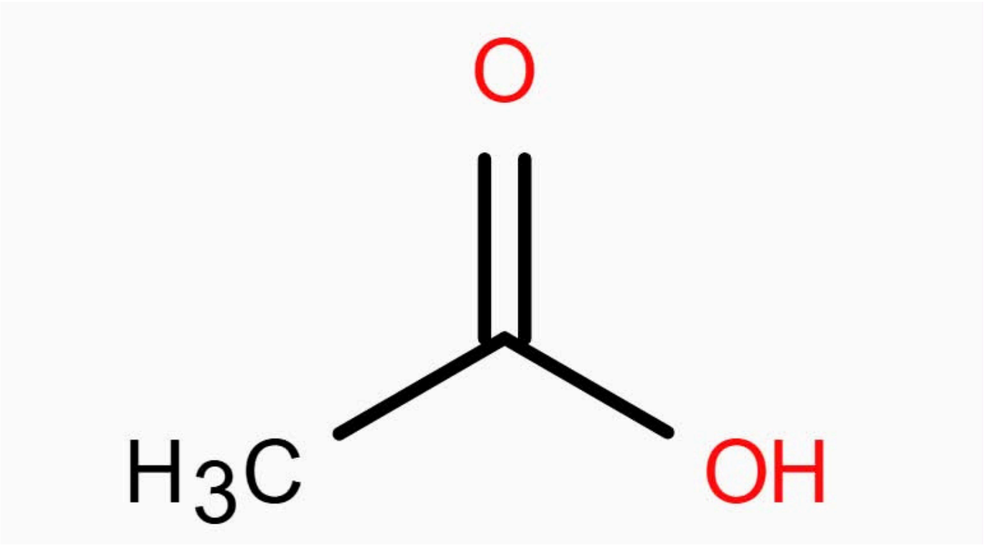
2.Condensed Structural Formula

CH_3COOH

3.Lewis Structure



4.Bond-Line Formula



CARD 04

Propanone (Acetone)

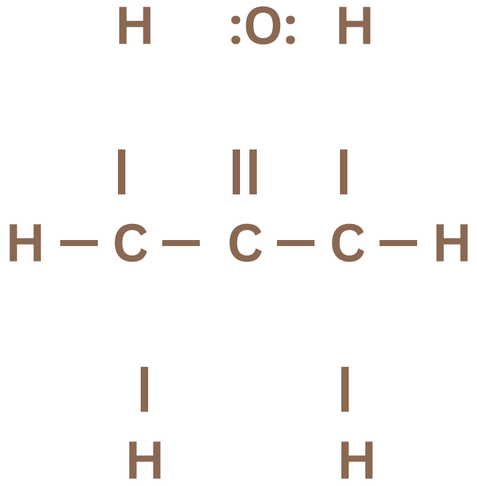
1.Molecular Formula

C_3H_6O

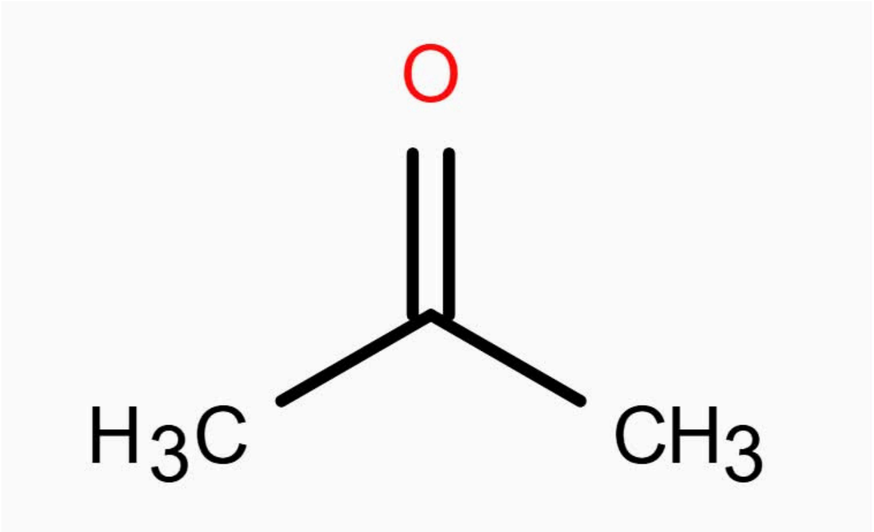
2.Condensed Structural Formula

CH_3COCH_3

3.Lewis Structure



4.Bond-Line Formula



CARD 06

Ethyl Acetate

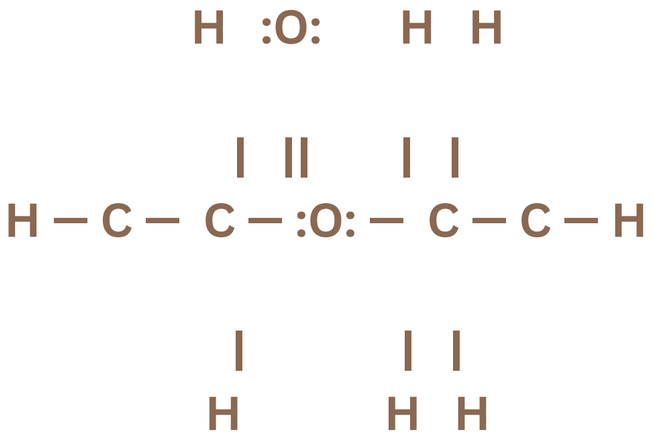
1.Molecular Formula

$C_4H_8O_2$

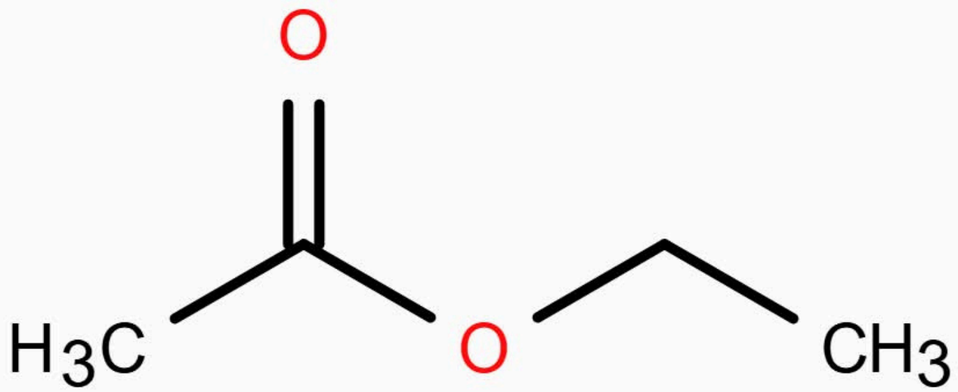
2.Condensed Structural Formula

$CH_3COOCH_2CH_3$

3.Lewis Structure



4.Bond-Line Formula



CARD 07

Cyclohexanone

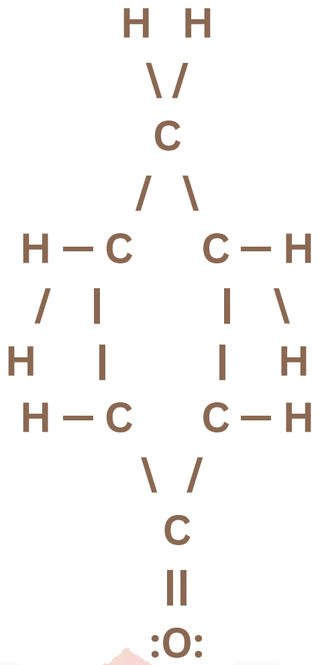
1.Molecular Formula

$C_6H_{10}O$

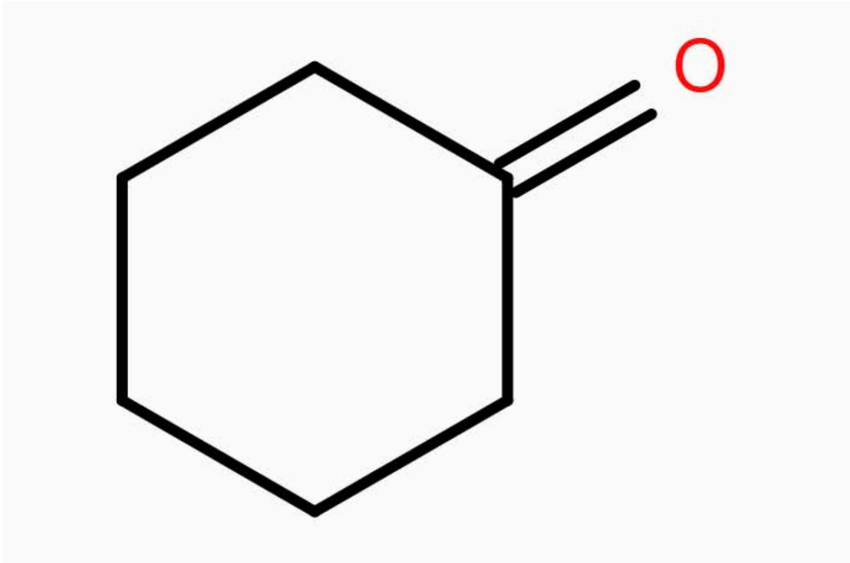
2.Condensed Structural Formula

$(CH_2)_5CO$

3.Lewis Structure



4.Bond-Line Formula



CARD 08

Aspirin

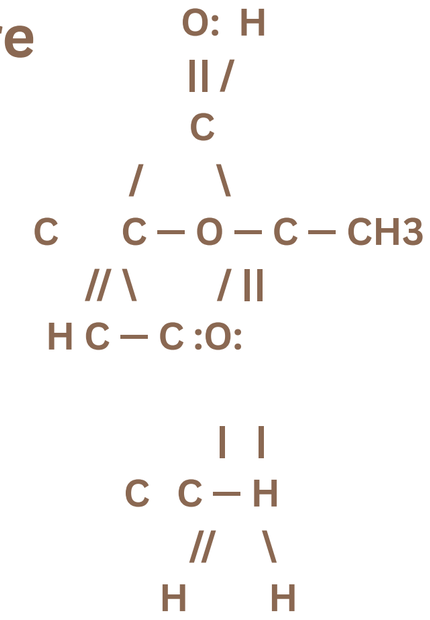
1.Molecular Formula

$C_9H_8O_4$

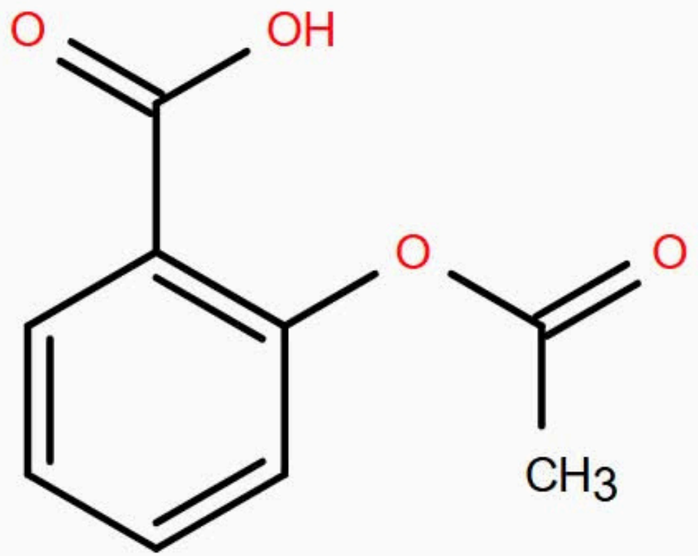
2.Condensed Structural Formula

$CH_3COOC_6H_4COOH$

3.Lewis Structure



4.Bond-Line Formula



CARD 09

Paracetamol

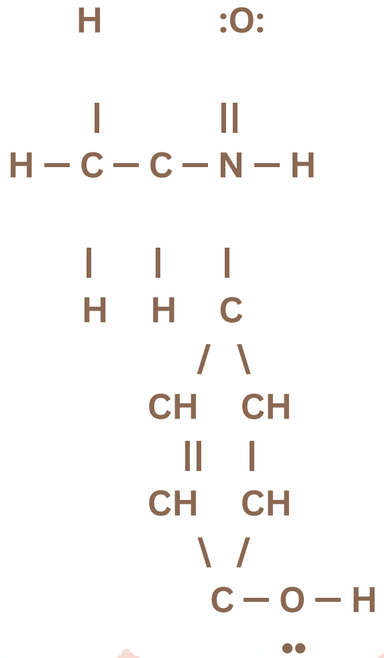
1.Molecular Formula

$C_8H_9NO_2$

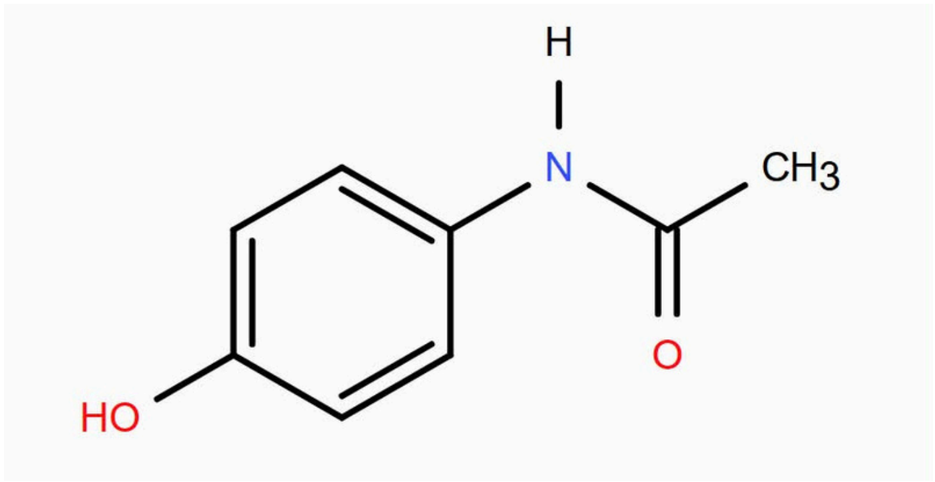
2.Condensed Structural Formula

$HOC_6H_4NHCOCH_3$

3.Lewis Structure



4.Bond-Line Formula



CARD 10

Caffeine

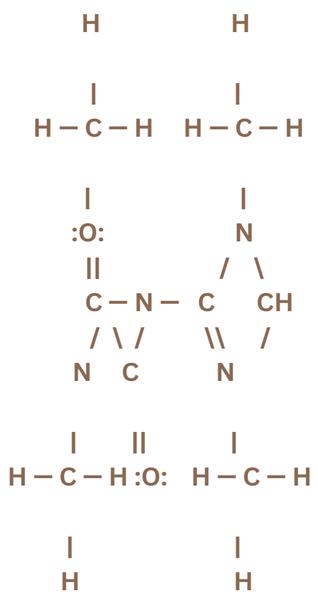
1.Molecular Formula

$C_8H_{10}N_4O_2$

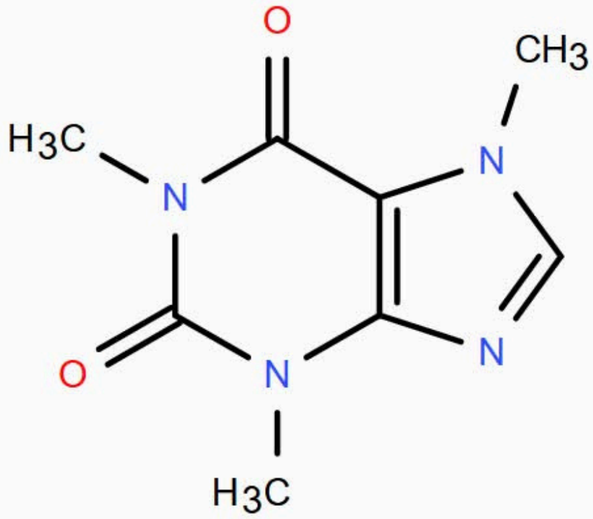
2.Condensed Structural Formula

$C_8H_{10}N_4O_2$

3.Lewis Structure



4.Bond-Line Formula



Thats All, Thank You!!!





